COOLING ENVIRONMENTS

ASHRAE-OCP joint PG25 mixing study *Update*

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Summary & Executive Agenda

Overview of the ASHRAE-OCP joint PG25 coolant mixing study ***update***

01

Goal & Approach

Addressing operational fluid mixing through standardized testing and collaboration.

02

Test Matrix

Evaluating 50:50 fluid mixes across chemical, corrosion, and material metrics.

03

Phased Plan

Structured 3-phase execution roadmap from lab screening to simulated service.

04

Phase 1 Focus

In-progress benchtop screening using ASTM D8040 and D471 methods.

05

Next Steps

Simulated-service testing loop validation planned for Nov-Dec '26.

CALL TO ACTION

Get Involved in establishing standardized simulated-service and cold plate testing methods

Join the OCP Cooling Environments project community to contribute test design and shape performance guidelines.

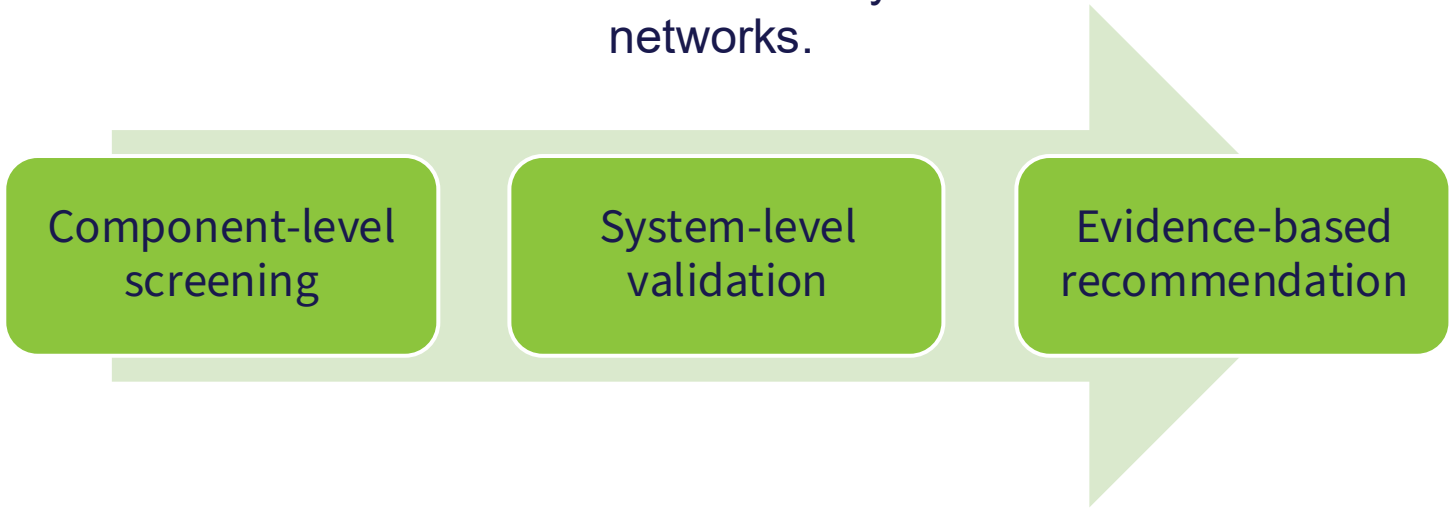
Goal and approach

Goal: To address coolant mixing and provide industry guidance, this initiative quantifies mixed coolant performance using established ASTM standards and new simulated-service testing methodologies.

Operational reality
PG25 fluids from different suppliers may be mixed intentionally or unintentionally in production during deployment and maintenance

Industry Gap
No standardized simulated-service method currently evaluates adverse effects from mixed coolants in secondary fluid networks.

Collaboration
ASHRAE-OCP is a platform for fluid suppliers, hyperscalers, chip and server vendors to develop a scientific test data from various perspectives



A phased approach reduces uncertainty before resources are committed to extended loop testing

Test matrix and performance metrics

Performance Assessment Metrics

- **Chemical compatibility**
  Additive interaction, instability, precipitation
- **Corrosion risk**
  Changes in corrosion tendency or inhibitor protection
- **Material compatibility**
  Response before/after EPDM exposure

Matrix of 50: 50 mixed fluids

Fluid	A	B	C	D
A	Unmixed	A-B / B-A	A-C / C-A	A-D / D-A
B		Unmixed	B-C / C-B	B-D / D-B
C			Unmixed	C-D / D-C
D				Unmixed

Mixed fluid samples are prepared by a third party

Testing sequence

Phase 1
Benchtop testing
ASTM D8040 and ASTM D471

→

Phase 2
Simulated service testing

→

Phase 3
Ad-hoc testing
e.g. cold plate autopsy

Phased study plan

Phase 1	Phase 2	Phase 3
Benchtop (lab) screening	**Simulated-service test**	**Follow-on scope**
ASTM testing of unmixed and mixed fluids, including fluid characterization before and after ASTM D8040 and ASTM D471	Liquid loop with thermal test vehicle and metal coupon rack, testing unmixed and mixed fluids at an agreed frequency	Scope will be defined based on Phase 1 and Phase 2 findings and remaining industry questions
Lead Fluid vendors	**Lead** Hyperscalers	**Lead** TBD
In progress	Planned	To be discussed
Target: Oct ‘26	Target: starting Nov-Dec ’26 for 6 month testing	Target: To be defined

Phase 1 - Benchtop (Lab) Screening

Leverage established ASTM testing methods to screen fluids prior to simulated-service testing

In progress

Benchtop screening: test methods

Analytical Methods

- **Elastomer Performance (*ASTM D471*)**
  - Peroxide cured EPDM
  - Volume, hardness, tensile strength, and elongation change
- **Metal Corrosion Performance (*ASTM D8040 - 18*)**
  - C11000 (Copper)
  - SS304 (Stainless Steel)
  - SS316 (Stainless Steel)
  - Brass
- **Pre- and Post-ASTM Method Fluid Analysis**
  - Test fluid evaluated via analytical panel before and after ASTM testing to assess potential failure modes

Fluid Analytical Methods

Property	Method	What potential risks it identifies
Appearance	Visual	Additive instability, phase separation, precipitation
pH	ASTM D1287	Corrosion, degradation
Reserve alkalinity	ASTM D1121	Future pH collapse and loss of corrosion protection
Phosphate	ASTM D5827	Incompatibility leading to precipitation
Anions	ASTM D5827	Increased corrosion potential
Degradation acids	ASTM D5827	Fluid degradation
Foaming	ASTM D1881	Reduced coolant performance
Turbidity	In-house	Additive fallout, EPDM swelling
Metals	ASTM D6130	Early corrosion indication, contamination
Inhibitors	In-house	Additive depletion and corrosion

Phase 1 Results

Phase 1 results will be available late October of 2026

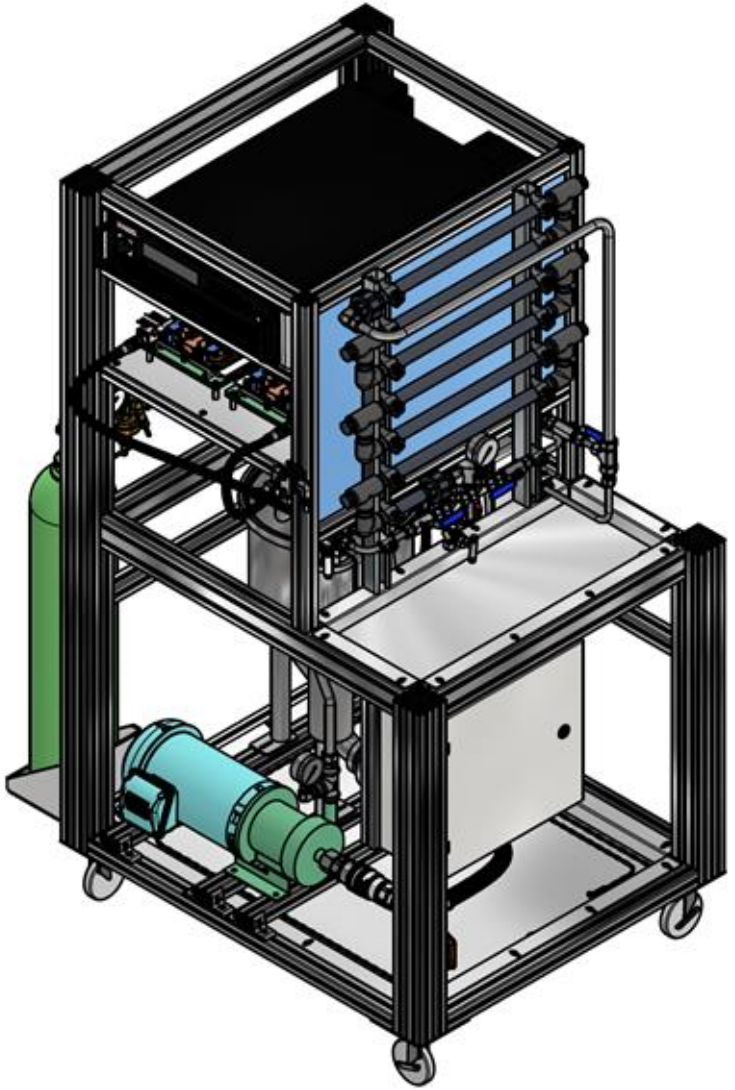
Phase 2 - Simulated-Service Testing

Establishing a simulated-service method to qualify data-center TCS coolants

Planned start – Nov of 26

Simulated-service: a TCS test-method gap

- **Lack of Existing Standards:** Direct liquid cooling currently lacks a standard simulated-service method comparable to ASTM D2570 for data-center TCS.
- **Phase 2 Pilot Objective:** Establishes a repeatable, system-level method for qualifying coolants and assessing surface mixing risk.
- **Chemistry-Agnostic Approach:** The design puts all TCS coolant families (PG, EG, water-based) on neutral footing, prioritizing the measurement method over preference for any specific chemistry.
- **Operational Scope:** The method incorporates startup and operation phases rather than excluding them.
- **Focus on Methodology:** This presentation introduces the testing framework rather than specific results; testing is scheduled to begin in November and run for six months.
- **New Industry Collaboration:** A new OCP workstream, the "Coolant Technologies Testing Workstream," is launching and will further develop standard testing methodologies.



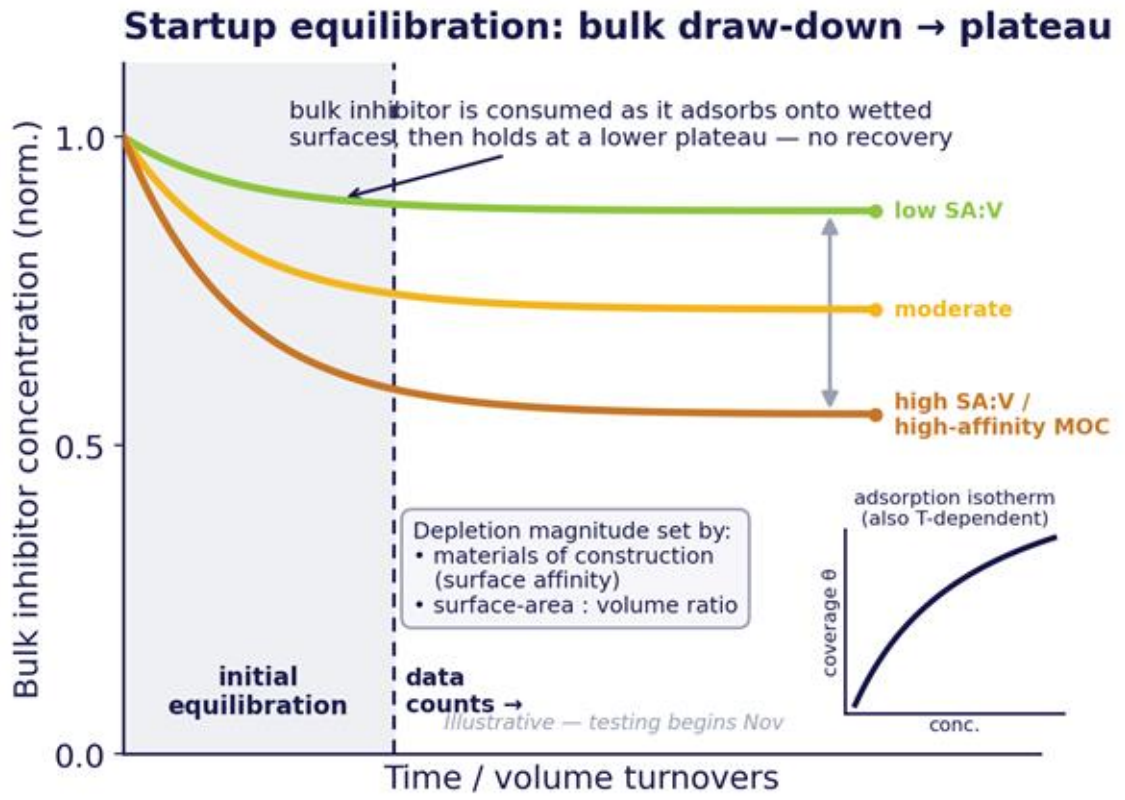
Building the method — start with procedure

To ensure consistent test results, we have formalized our system flushing and filling procedures:

- **Standardized Velocities and Volumes:** Mandated protocols establish specific flushing velocities and durations normalized to the system volume.
- **Clean Initial Conditions:** Chemical addition is restricted to surfactants and biocides to guarantee repeatable initial conditions, regardless of the coolant type.
- **No Additional Corrosion Inhibitors:** Corrosion inhibitors are intentionally excluded during startup to avoid bias, ensuring the evaluation focuses solely on the coolant's inhibitory characteristics rather than the startup chemistry.
- **Baseline Water Quality:** All flush water strictly adheres to well defined standards to establish a rigorous baseline.

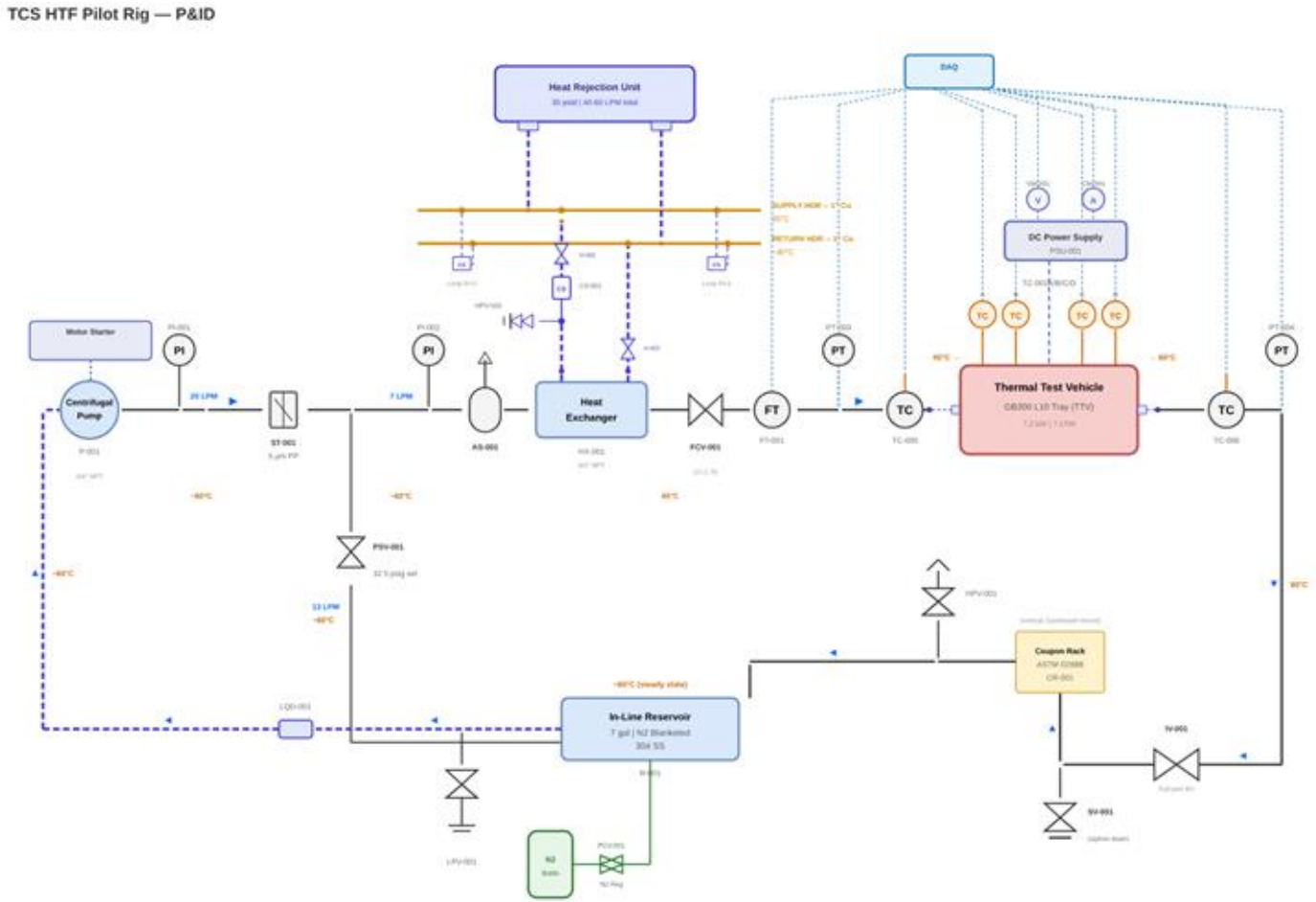
Startup in action: the equilibration step

- **Significance of Startup Procedures:** Establishing correct startup protocols is critical for accurate testing outcomes.
- **Mechanism of Inhibitors:** Inhibitors work by adsorbing a protective film onto metal surfaces, with coverage dictated by concentration- and temperature-dependent adsorption isotherms.
- **Initial Sorption Impact:** When fresh coolant is introduced, initial surface sorption causes a depletion of the bulk inhibitor (an "early dip"), the magnitude of which correlates with the system's volume-to-wetted-area ratio and material.
- **Procedural Requirement:** To ensure valid data, we chose to include a deliberate initial equilibration period, followed by a flush and refill with unbiased coolant.
- **Standardization and Precedent:** Similar approach found in ASTM D2570, which serves as a precedent for the TCS analog.



The loop — the repeatable unit

- **Modular Testing Platform Design**
  - Engineered as an instrumented, repeatable unit.
  - Architecture is adaptable to support evolving testing methodologies.
- **Robust Material Selection**
  - System components use corrosion-resistant materials to isolate test surfaces (*i.e.*, 316-SS, EPDM-Perox, PTFE, *etc.*).
  - Exception: The coalescing air separator is made of chrome-plated brass.
- **Simplified Control Strategy**
  - Flow is regulated by a fixed-speed pump, control valve, and back-pressure system.



Reservoir, not an expansion tank

Design Standardization

Standardizing designs is essential to consolidate industry knowledge and mitigate errors.

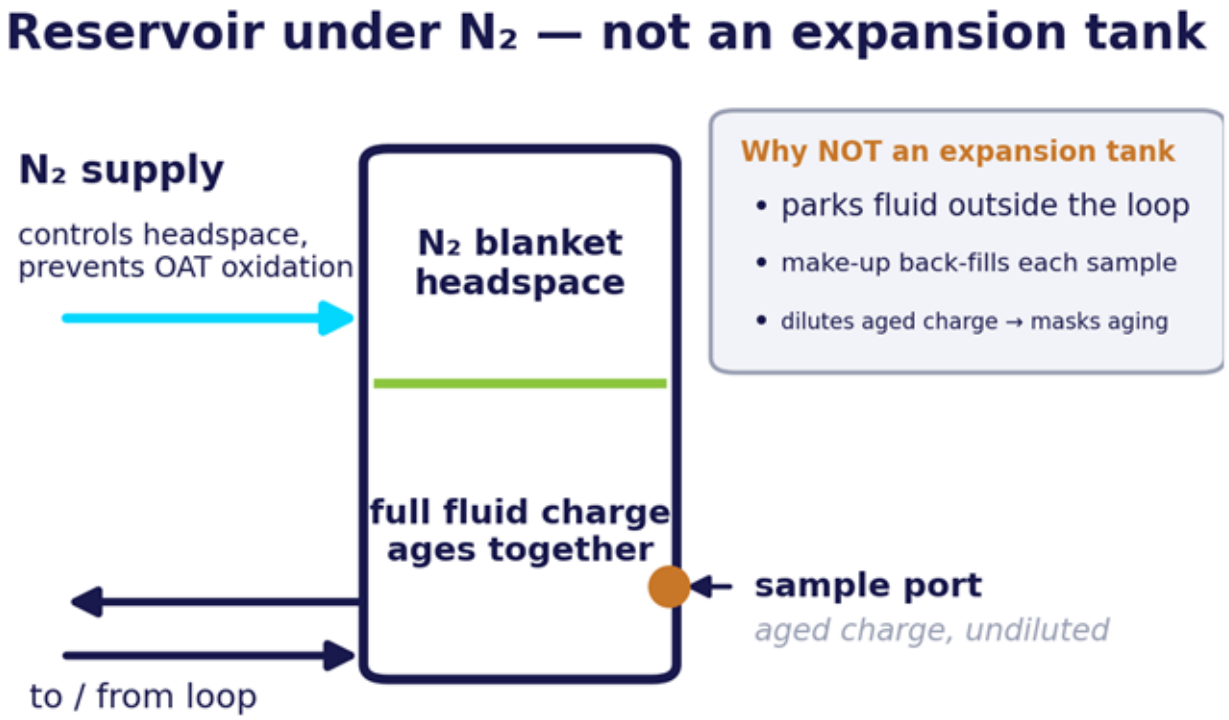
Fluid Management: Reservoirs vs. Expansion Tanks

- **Reservoirs:** Maintain the entire fluid charge within the loop, ensuring uniform aging and simpler monitoring.
- **Expansion Tanks:** Isolate fluid, complicating monitoring and risking measurement inaccuracy if fresh fluid dilutes the aged sample.

Controlling Oxidation

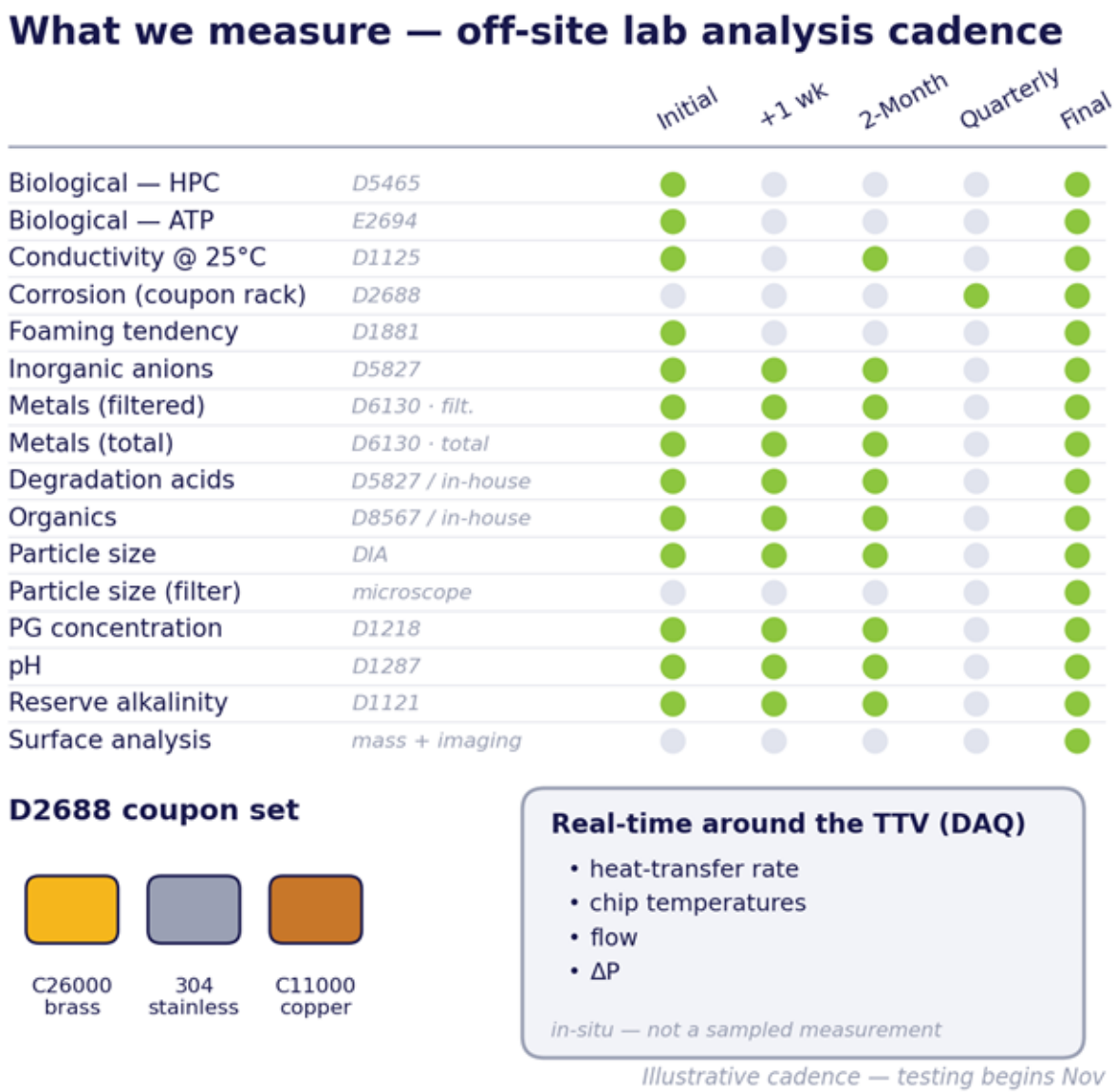
A nitrogen (N_2) blanket is applied to control the reservoir headspace to prevent oxygen/air ingress.

OAT oxidation risk



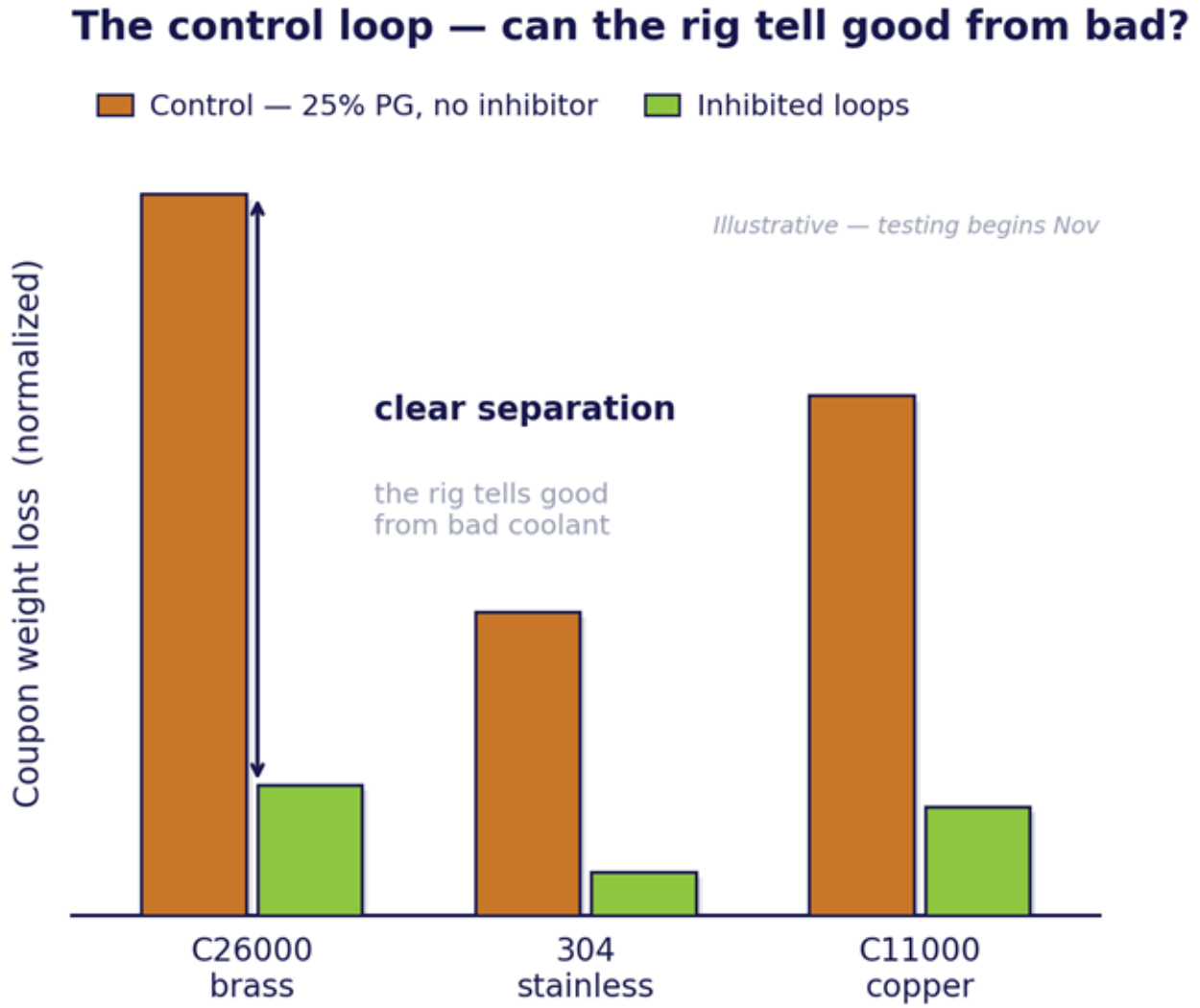
What we measure

- **Corrosion Qualification Metric:** Utilizes ASTM D2688 weight-loss coupons, specifically testing C26000 brass, 304 stainless steel, and C11000 copper.
- **Coupon Material:** Addresses both curated IT rack materials (*e.g.*, C11000, 304-SS) and facility-side materials (*e.g.*, C26000 cartridge brass ~ 30% Zn).
- **Periodic & Continuous Monitoring:**
  - Fluid analysis performed on a set cadence and leverages standard methods to the greatest extent possible.
  - Real-time performance monitoring around the TTV (Thermal Test Vehicle) via data acquisition (DAQ).
- **Cold-Plate Assessment:** Teardowns are available post-test; however, in-situ monitoring relies on temperature and pressure data rather than coupon sampling, as D2688 does not capture conditions under active heat transfer.



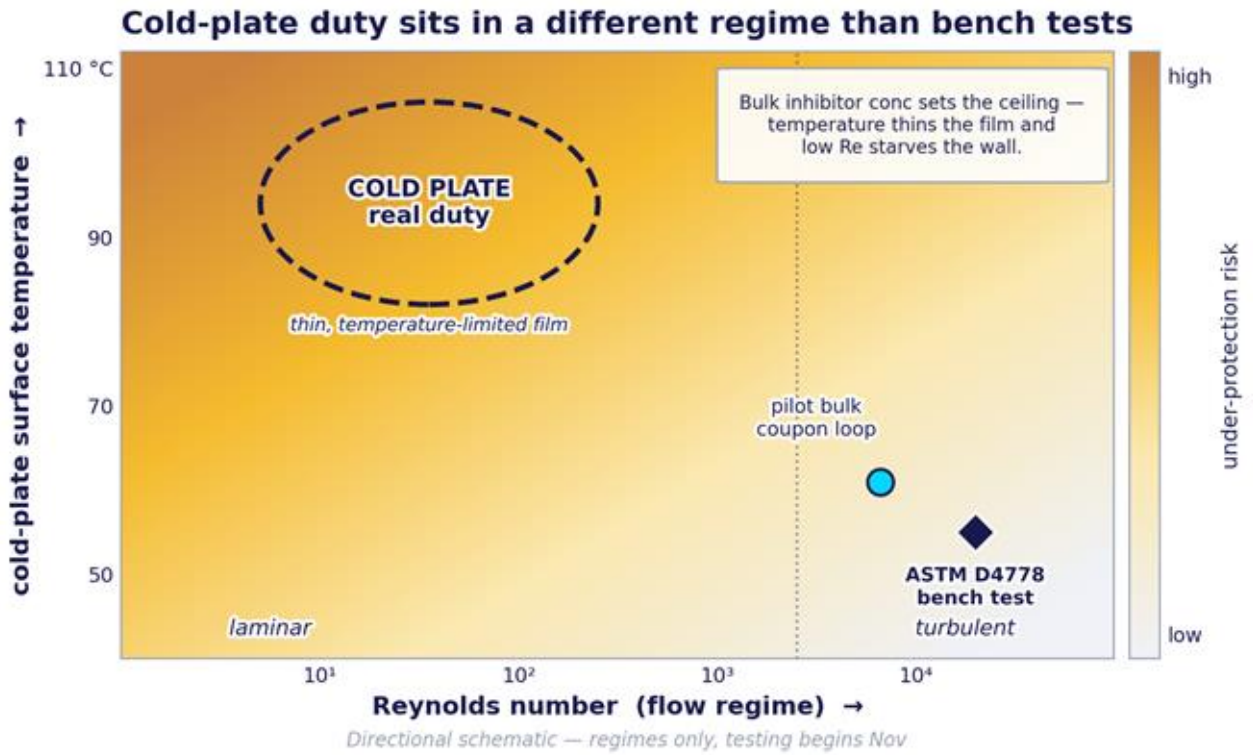
The control: can the rig tell good from bad?

- Our initial pilot methodology leverages a 25% propylene glycol (PG) solution without corrosion inhibitors as a control.
- Compare these control results against data from inhibited loops to evaluate the proposed method's ability to distinguish between different coolant types.
- Use any failure to differentiate between coolants as feedback for future test design iterations.



Questions we can't yet answer — and why

- **Challenge:** Current methods cannot accurately measure coolant corrosivity or inhibitor performance within cold-plate microchannels under real heat flux conditions.
- **Limitation of Existing Standards:**
  - ASTM D2688 is designed for isothermal environments, failing to account for the impact of heat transfer on surface film formation.
  - ASTM D4778 utilizes turbulent heated tubes, which does not effectively model the laminar flow regimes of cold plates.
- **Current Reality:** Corrosion assessment in cold plates currently relies solely on destructive teardown.
- **The Opportunity:** Develop a standardized, cold-plate-representative coupon rack—effectively a "D4778 for cold plates"—designed to utilize a new testing rig once the standard is established.



Call to action: build the standards

The industry needs two things that don't exist yet:

1. A TCS-centric simulated-service method — the D2570 analog for data-center cooling.
2. A samplable heat-transfer corrosion method for laminar cold plates — “a D4778 for cold plates.”

Contact the project leads for more info



Know before you flow.

Jake Davis · Meta — with Sung A Kim · Castrol

OCP Global Summit 2026

Thank You!

